

Notes: Peress & Schmidt (JFE, 2024)

Uncertainty about what is in the price

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1 Big picture

- **Question.** When a speculator has a private signal, how does she know whether the signal is genuinely new or already reflected in recent prices?
- **Core concept.** The paper studies “uncertainty about what is in the price” (UWIP): informed investors are uncertain about whether their signal is stale news or novel information.
- **Main theoretical mechanism.** Speculators use recent price movements to infer the novelty of their signals. A positive signal after a recent price increase may be stale; a positive signal after a recent price decline is more likely to be novel.
- **Market-maker implication.** Market makers understand this inference. After recent price increases, buy volume is less informative because it may come from stale positive news, while sell volume is more informative. After recent price decreases, the logic reverses.
- **Distinct prediction 1.** Return skewness is negatively related to lagged returns. After price increases, price updates are more left-skewed; after price decreases, they are more right-skewed.
- **Distinct prediction 2.** Price impact depends asymmetrically on past returns: for net buys it decreases with past returns, and for net sells it increases with past returns.
- **Empirical setting.** The authors use an exhaustive sample of NYSE-traded stocks from 1993 to 2014, with CRSP daily data and intraday TAQ data.
- **Skewness evidence.** Daily realized skewness from five-minute returns is strongly negatively related to lagged returns. A one-standard-deviation increase in lagged 10-day returns lowers skewness by about 9% of its standard deviation.
- **Price-impact evidence.** On net-buy days, price impact is negatively related to past returns; on net-sell days, price impact is positively related to past returns. With a 10-day lookback, this creates a buy-versus-sell wedge around 15% of a standard deviation for price impact costs.
- **Information environment tests.** The effects weaken after earnings announcements, for larger firms, and for firms with more analyst coverage. They strengthen for firms that operate in many business segments relative to their size.
- **Order-flow source test.** The price-impact asymmetry is strong for institutional order imbalance but much weaker for retail order imbalance.
- **Alternative mechanisms.** The results are not well explained by short-sale constraints, inventory risk, or S&P 500 index effects.
- **Economic takeaway.** Prices are informative, but investors do not know exactly *which* information they contain. This uncertainty changes trading aggressiveness, price impact, return skewness, and price informativeness.

2 Data and measurement (returns, order flow, skewness, and price impact)

2.1 Why the empirical setting is useful

- The paper studies common stocks traded on the NYSE from 1993 to 2014.
- The sample combines CRSP daily stock data with intraday TAQ trades and quotes.
- The setting is ideal because the model makes predictions about high-frequency return distributions and trade-direction-specific price impact.
- NYSE stocks provide a broad cross-section of liquidity, information environments, analyst coverage, size, and trading activity.

Interpretation. The empirical strategy is not just testing whether past returns predict future returns. It tests whether past returns change the asymmetry of price updates and the informational content of buy versus sell order flow.

2.2 Sample construction

- The sample is the union of CRSP and TAQ over 1993–2014.
- The authors focus on common stocks, share codes 10 and 11.
- Penny stocks with closing prices below \$1 are excluded.
- TAQ quotes are filtered and adjusted using the Holden and Jacobsen method for withdrawn or canceled quotes and interpolated quote times.
- Trades are signed using the Lee and Ready algorithm.

Interpretation. The goal is to create stock-day and intraday measures that are comparable across a long period with many stocks.

2.3 Trade imbalance

Let signed trade size be positive for buys and negative for sells. The daily share trade imbalance is the sum of signed shares over the trading day. The dollar trade imbalance is the sum of signed trades multiplied by the prevailing mid-quote:

$$\text{TradeImbalance}_{it} = \sum_{\tau \in t} \text{Sign}_{i\tau} \times \text{Shares}_{i\tau} \times \text{Midquote}_{i\tau}.$$

Interpretation. The sign of the daily imbalance splits the sample into net-buy and net-sell days. The model's distinctive prediction is about how price impact differs across these two subsamples after prior price moves.

2.4 Realized return skewness

Daily return skewness is measured from five-minute midpoint returns:

$$\text{Skewness}_{it} = \frac{\sqrt{K} \sum_{k=1}^K (r_{itk} - \bar{r}_{it})^3}{\left[\sum_{k=1}^K (r_{itk} - \bar{r}_{it})^2 \right]^{3/2}}.$$

- r_{itk} is stock i 's midpoint return in five-minute interval k on day t .

- $\bar{r}_{it}$ is the average five-minute return that day.
- Negative values indicate a fatter left tail; positive values indicate a fatter right tail.

Interpretation. This is the daily object closest to the model’s prediction that the return distribution becomes left-skewed after price increases and right-skewed after price decreases.

2.5 Price impact costs

The first price-impact measure is a signed Amihud-style ratio:

$$\text{PriceImpact}_{it} = \frac{\text{Return}_{it}^{adj}}{\text{TradeImbalance}_{it}}.$$

- The return is adjusted for autocorrelation using lagged daily returns.
- The denominator is signed dollar trade imbalance.
- The measure asks how much the stock price moves for one dollar of net buying or net selling.

Interpretation. A higher value means a larger price response to signed order flow. The model predicts that this value should fall with past returns on net-buy days and rise with past returns on net-sell days.

2.6 Additional price-impact measures

The paper uses three additional adverse-selection price-impact measures:

- **Lambda:** the slope from regressions of five-minute stock returns on signed order flow.
- **Quote-based price impact:** the dollar-weighted percentage change in the midquote from before a transaction to five minutes after it.
- **Ln(Amihud):** the log of the standard Amihud illiquidity ratio, based on absolute return divided by dollar volume.

Interpretation. Using four measures reduces the concern that the results are an artifact of one liquidity proxy.

2.7 Past returns and controls

- Past returns are measured over one, five, and ten trading days.
- The main control variables are past turnover and past squared return over the same lookback window.
- Regressions include stock-month fixed effects and date fixed effects.
- Standard errors are double-clustered by stock and day.

Interpretation. Stock-month fixed effects absorb persistent stock-level liquidity, volatility, size, and information characteristics; date fixed effects absorb market-wide conditions.

2.8 Information environment variables

The auxiliary tests use variables that proxy for the ease of knowing what is already reflected in prices:

- an earnings announcement dummy for whether the firm had an announcement over the past ten trading days;
- analyst coverage, measured as log one plus the number of analysts;

- market capitalization;
- a complicated-firm measure: the residual number of business segments after removing the effect of size;
- S&P 500 additions and deletions;
- short-selling fees from IHS Markit.

Interpretation. UWIP should be lower when public attention is high and information is easier to interpret. It should be higher for complicated firms.

2.9 Retail and institutional order flow

- Retail buys and sells are identified using the Boehmer et al. method, based on subpenny price improvement in TAQ.
- Retail order-flow data are available from 2005 to 2014.
- Institutional imbalance is defined as total imbalance minus retail imbalance.
- The price-impact regressions are repeated after splitting days by institutional or retail imbalance.

Interpretation. If the effect is about informed trading and adverse selection, the asymmetry should be stronger for institutional flow than for retail flow.

3 Models and empirical specifications (all equations plus interpretation)

3.1 Model setup

There are three dates, $t = 0, 1, 2$. The payoff is

$$\theta = \theta_1 + \theta_2, \quad \theta_j \in \{+\sigma, -\sigma\}.$$

At $t = 0$, market makers observe one component θ_m , where $m \in \{1, 2\}$, and set

$$p_0 = E(\theta \mid \theta_m) = \theta_m.$$

At $t = 1$, speculators observe θ_s , where $s \in \{1, 2\}$, and submit market orders. Noise traders submit $n \sim U[-1, 1]$. Market makers observe total order flow ω and set

$$p_1 = E(\theta \mid \theta_m, \omega).$$

Interpretation. The uncertainty is not about whether the speculator sees a signal. It is about whether the signal is the same component already observed and priced by the market maker.

3.2 Benchmark 1: both m and s are common knowledge

If $m = s$, speculators have no information advantage and do not trade. If $m \neq s$, speculators know their information is novel and trade

$$x^{(1)} = \sqrt{\frac{1}{\gamma\sigma}}.$$

Market makers set the price change symmetrically:

$$p_1 - p_0 = \begin{cases} +\sigma, & -x^{(1)} + 1 < \omega \leq x^{(1)} + 1, \\ 0, & x^{(1)} - 1 \leq \omega \leq -x^{(1)} + 1, \\ -\sigma, & -x^{(1)} - 1 \leq \omega < x^{(1)} - 1. \end{cases}$$

Interpretation. Price impact does not depend on the prior price. Buy and sell orders are symmetric.

3.3 Benchmark 2: only m is common knowledge

Speculators know whether their signal is novel, but market makers are unsure whether informed speculators are present. The price function has more regions, but the key property remains:

$$p_1 - p_0 \quad \text{does not depend on the realization of } p_0.$$

Interpretation. Market makers face uncertainty about whether informed traders exist, but speculators do not face UWIP. This alone does not produce the model's central asymmetry.

3.4 UWIP equilibrium: neither m nor s are common knowledge

Speculators know their signal and observe p_0 , but they do not know which component produced p_0 .

- If $\theta_m \neq \theta_s$, the signal must be novel. Speculators trade aggressively, with size $x^{(3)}$.

- If $\theta_m = \theta_s$, the signal may be stale. Speculators trade less aggressively, with size $y^{(3)}$.
- In equilibrium:

$$y^{(3)} = \frac{2 - \gamma\sigma(x^{(3)})^2}{\gamma\sigma x^{(3)}} \leq x^{(3)}.$$

Interpretation. Speculators are more cautious when their signal confirms the recent price move because it may already be in the price.

3.5 UWIP price function after prior positive news

If $\theta_m = +\sigma$, then

$$p_1 - p_0 = \begin{cases} +\frac{1}{3}\sigma, & -x^{(3)} + 1 < \omega \leq y^{(3)} + 1, \\ 0, & y^{(3)} - 1 \leq \omega \leq -x^{(3)} + 1, \\ -\sigma, & -x^{(3)} - 1 \leq \omega < y^{(3)} - 1. \end{cases}$$

Interpretation. After a price increase, buy volume may be stale positive-news trading and receives only a small price impact. Sell volume contradicts the earlier price increase and is more informative, so the downward price impact is larger.

3.6 UWIP price function after prior negative news

If $\theta_m = -\sigma$, then

$$p_1 - p_0 = \begin{cases} +\sigma, & -y^{(3)} + 1 < \omega \leq x^{(3)} + 1, \\ 0, & x^{(3)} - 1 \leq \omega \leq -y^{(3)} + 1, \\ -\frac{1}{3}\sigma, & -y^{(3)} - 1 \leq \omega < x^{(3)} - 1. \end{cases}$$

Interpretation. After a price decrease, sell volume may be stale negative-news trading and has lower price impact. Buy volume contradicts the earlier decline and is more informative.

3.7 Price informativeness

The paper defines price informativeness as

$$PI \equiv Var(E(\theta \mid p_1, p_0)).$$

For the three equilibria:

$$PI^{(1)} = \sigma^2 \left(1 + \frac{1}{2}x^{(1)}\right), \quad PI^{(2)} = \sigma^2 \left(1 + \frac{1}{3}x^{(2)}\right), \quad PI^{(3)} = \sigma^2 \left(1 + \frac{1}{6}(x^{(3)} + y^{(3)})\right).$$

The UWIP equilibrium satisfies

$$PI^{(3)} < PI^{(1)}, \quad PI^{(3)} < PI^{(2)}.$$

Interpretation. UWIP lowers price informativeness because speculators trade less aggressively when they fear their signals may be stale.

3.8 Hypothesis 1: skewness and past returns

The skewness regression is

$$\text{Skewness}_{it} = \alpha_{i\tau} + \alpha_t + \beta \text{PastReturn}_{it} + \gamma' X_{it-1} + \varepsilon_{it}.$$

The model predicts

$$\beta < 0.$$

Interpretation. The coefficient tests whether return distributions become more left-skewed after price increases and more right-skewed after price decreases.

3.9 Hypothesis 2: price impact and past returns

The price-impact regressions are run separately by net-buy and net-sell days:

$$\text{PriceImpact}_{it} = \alpha_{i\tau} + \alpha_t + \beta^b \text{PastReturn}_{it} + \gamma' X_{it-1} + \varepsilon_{it}, \quad \text{if TradeImbalance}_{it} > 0,$$

$$\text{PriceImpact}_{it} = \alpha_{i\tau} + \alpha_t + \beta^s \text{PastReturn}_{it} + \gamma' X_{it-1} + \varepsilon_{it}, \quad \text{if TradeImbalance}_{it} < 0.$$

The model predicts

$$\beta^b < 0, \quad \beta^s > 0.$$

Interpretation. Buy impact should be lower after positive past returns because buys may be stale-news trading. Sell impact should be higher after positive past returns because sells contradict the earlier price move.

3.10 Information-environment interactions

The earnings-announcement test augments the skewness regression:

$$\text{Skewness}_{it} = \alpha_{i\tau} + \alpha_t + \beta_1 \text{PastReturn}_{it} + \beta_2 EA_{it} + \beta_3 (\text{PastReturn}_{it} \times EA_{it}) + \gamma' X_{it-1} + \varepsilon_{it}.$$

The model predicts

$$\beta_1 < 0, \quad \beta_3 > 0.$$

The analogous price-impact regressions predict:

$$\beta_1^b < 0, \quad \beta_1^s > 0, \quad \beta_3^b > 0, \quad \beta_3^s < 0.$$

Interpretation. Earnings announcements make it easier to know what information caused recent price moves, so UWIP and its empirical footprint should weaken.

3.11 Retail versus institutional flow

The price-impact regressions are repeated after replacing total trade imbalance with institutional imbalance or retail imbalance.

$$\text{InstitutionalImbalance}_{it} = \text{TotalImbalance}_{it} - \text{RetailImbalance}_{it}.$$

Interpretation. The model is about adverse selection from potentially informed speculators. If institutions are more likely to be informed, the asymmetry should be stronger for institutional flow.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the main skewness effect

- Identification comes from within-stock-month variation in past returns and realized daily skewness.
- Stock-month fixed effects absorb persistent stock characteristics.
- Date fixed effects absorb market-wide shocks.
- Controls include past turnover and past volatility over the same lookback window.

Interpretation. The skewness result is not a cross-sectional statement that some stocks are more skewed than others. It is a within-stock statement about how the next day's return distribution changes after recent price moves.

4.2 What identifies the price-impact asymmetry

- Price-impact regressions are split into net-buy and net-sell days.
- The coefficient on past return is estimated separately in the two subsamples.
- The model predicts opposite signs in the two subsamples.
- This sign pattern is difficult to explain with a generic liquidity trend.

Interpretation. The buy-sell asymmetry is the central empirical footprint of UWIP.

4.3 Why the information-environment tests matter

- UWIP should be lower after earnings announcements because the public news source is clear.
- UWIP should be lower for firms with more analyst coverage and for larger firms.
- UWIP should be higher for complicated firms with more business segments relative to size.
- The paper finds these patterns.

Interpretation. These tests make the mechanism more credible because they show that the footprint strengthens and weakens where UWIP should be high or low.

4.4 Why the institutional-versus-retail test matters

- The asymmetry should be stronger when market makers fear adverse selection.
- Institutional orders are more likely to be informed than retail orders.
- The paper finds strong asymmetry for institutional order imbalance and weaker asymmetry for retail imbalance.

Interpretation. This connects the empirical result to informed trading rather than to uninformed order-flow pressure.

4.5 Addressing inventory risk

- Inventory-risk theories often predict that same-direction order flow should have higher price impact when it pushes market makers away from target inventory.
- UWIP predicts the opposite for the adverse-selection component: same-direction order flow can be stale-news trading and therefore less informative.
- The paper shows robustness to controlling for past trade imbalance and past spread measures.

Interpretation. The UWIP channel is not simply a repackaged inventory-cost channel.

4.6 Addressing short-sale constraints

- Some skewness theories rely on short-sale constraints delaying the incorporation of bad news.
- The paper uses stock lending fees as a proxy for tight short-sale constraints.
- The interaction of past returns with short fees does not strengthen the skewness-return relation.
- For price impact, short-sale constraints do not strengthen the buy-sell asymmetry in the predicted way.

Interpretation. The daily skewness and price-impact results do not appear to be driven by short-sale constraints.

4.7 What the paper cannot fully prove

- The empirical tests do not observe the exact private signals held by speculators.
- Trade signing by Lee and Ready is an inference, not direct order-origin information.
- Retail identification is available only for the 2005–2014 subperiod.
- UWIP may coexist with inventory risk, dynamic speculation, short-sale constraints, and attention limits.
- The expected-return, volatility, and price-informativeness predictions are harder to isolate empirically.

Interpretation. The paper’s strongest claim is that a distinctive pattern in skewness and price impact is consistent with UWIP and difficult to explain jointly with existing alternatives.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* format and places adjacent visuals together where possible to save space. The figures and tables are direct crops from the original paper.

5.1 Figure 1 and Figure 2: model setup and asymmetric price function

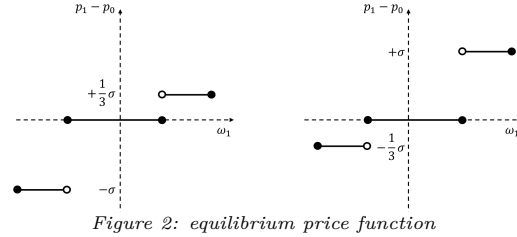
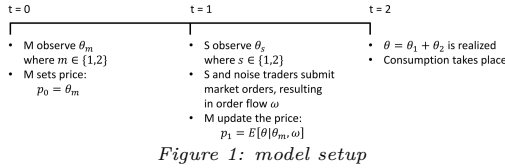
Read: Figure 1:

- At $t = 0$, market makers observe θ_m and set $p_0 = \theta_m$.
- At $t = 1$, speculators observe θ_s , submit market orders, and noise traders also trade.
- Market makers observe net order flow and set $p_1 = E[\theta \mid \theta_m, \omega]$.
- At $t = 2$, the payoff $\theta = \theta_1 + \theta_2$ is realized.

Read: Figure 2:

- Panel A shows the price function after prior positive news.
- After a prior price increase, buy order flow raises the price only by $+\sigma/3$, while sell order flow can lower the price by $-\sigma$.
- Panel B shows the reverse after prior negative news.
- This is the visual representation of asymmetric price impact under UWIP.

Economic message. The model turns past price movements into a filter for signal novelty. Order flow that agrees with the prior price move may be stale and has lower price impact; order flow that contradicts the prior move is more informative.



5.2 Table 1 and Table 2: descriptive statistics and baseline skewness tests

Read: Table 1:

- The overall sample has 22.43 million stock-day observations.
- Overall return skewness has mean -0.0152 , median 0.0309 , and standard deviation 2.8967 .
- Net-buy days have positive average skewness, 0.4756 .
- Net-sell days have negative average skewness, -0.4843 .
- Price impact costs are slightly higher on net-sell days than net-buy days, so the main result is not simply that buys are more costly than sells.

Read: Table 2:

- In the full sample, past-return coefficients are negative for all lookback windows: -0.8531 , -2.0598 , and -2.3196 .
- The 10-day coefficient implies an economically meaningful decline in skewness of about 9% of a standard deviation for a one-standard-deviation increase in past returns.

- The negative relationship is especially strong after negative past returns.
- The 5-day and 10-day positive-return subsamples also show negative coefficients.

Economic message. Table 2 is the first main empirical result: the daily return distribution becomes more left-skewed after prior price increases and more right-skewed after prior price declines.

Table 1
Descriptive statistics

Panel (a): Descriptive statistics for overall sample

	Mean	Median	Standard deviation
<i>Dependent variables</i>			
Return skewness	-0.0152	0.0309	2.8967
Price impact costs	0.6063	0.0087	5.8523
Lambda	0.1413	0.0224	0.4777
Quote-based price impact	0.3966	0.1443	0.7834
Ln(Amihud)	-16.9706	-17.8151	1.8021
<i>Independent variables</i>			
Past 1-day return	0.0008	0.0000	0.0348
Past 5-days return	0.0045	0.0016	0.0752
Past 10-days return	0.0090	0.0051	0.1055
Past 1-day turnover	0.0068	0.0036	0.0095
Past 5-days turnover	0.0070	0.0040	0.0087
Past 10-days turnover	0.0070	0.0042	0.0084
Past 1-day volatility	0.0013	0.0002	0.0035
Past 5-days volatility	0.0015	0.0005	0.0029
Past 10-days volatility	0.0015	0.0006	0.0027
N	22,433,401		

Panel (b): Descriptive statistics for net buy-days

	Mean	Median	Standard deviation
Return skewness	0.4756	0.2233	2.6732
Price impact costs	0.4923	0.0045	5.0265
Lambda	0.1254	0.0194	0.4469
Quote-based price impact	0.3739	0.1373	0.7434
Ln(Amihud)	-17.1294	-17.9773	1.7198
N	10,908,396		

Panel (c): Descriptive statistics for net sell-days

	Mean	Median	Standard deviation
Return skewness	-0.4843	-0.1869	3.0054
Price impact costs	0.6671	0.0147	5.0667

Table 1: descriptive statistics

Table 2

Return skewness and past returns.

	Lagged 1-day		Lagged 5-days		Lagged 10-days	
	All days (1)	≤ 0 (2)	All days (4)	≤ 0 (5)	All days (7)	≤ 0 (8)
Past return	-0.8531*** (-24.61)	-1.4637*** (-15.19)	0.6213** (6.34)	-2.0598*** (-74.27)	-3.1597*** (-71.72)	-2.3196*** (-74.27)
Past turnover	-0.2307 (-1.63)	1.8332*** (9.92)	-3.8769*** (-21.71)	-2.2566*** (-10.83)	-3.0702*** (-10.14)	-2.9107*** (-11.16)
Past volatility	12.5636*** (34.66)	18.2397*** (24.08)	-2.5886*** (-3.96)	20.4112*** (32.94)	6.3265*** (6.72)	30.1649*** (36.53)
t-stat of difference in past return coef.		(15.52)***		(17.53)***		(10.46)***
Stock-month FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
N	23,925,441	13,011,833	10,862,693	23,916,920	11,255,620	12,586,601
adj. R ²	0.06	0.06	0.08	0.06	0.08	0.07

This table reports the results from regressing daily return skewness on lagged raw returns as explained in Section 3.2. Columns (1)–(3) show the results for using lag returns as the key independent variable. Columns (4)–(6) show the results for using lagged cumulated 5-day returns as the key independent variable. Columns (7)–(9) show the results for using lagged cumulated 10-day returns as the key independent variable. The test statistics and p-values are shown in parentheses. Columns (1)–(3) are for net buy-days, and columns (4)–(6) are for net sell-days.

Table 2: return skewness and past returns

5.3 Table 3: price impact and past returns

Read:

- Panel A uses price impact costs. Past-return coefficients are negative on net-buy days and positive on net-sell days for all lookback windows.
- With a 10-day lookback, the coefficient is -4.5070 for net buys and $+3.9062$ for net sells.
- Panel B uses lambda. The 10-day coefficients are -0.1563 for net buys and $+0.0517$ for net sells.
- Panel C uses quote-based price impact. The 10-day coefficients are -0.2752 for net buys and $+0.1085$ for net sells.
- Panel D uses $\ln(\text{Amihud})$. The 10-day coefficients are -1.0076 for net buys and $+0.3665$ for net sells.
- The signs are strongly consistent across measures and horizons.

Economic message. This is the paper's central price-impact evidence. Buy order flow has lower impact after positive past returns; sell order flow has higher impact after positive past returns.

Table 3

Price impact and past returns.

Panel (a): Dependent variable: Price impact costs

	Lagged 1-day		Lagged 5-days		Lagged 10-days	
	Net buys (1)	Net sells (2)	Net buys (3)	Net sells (4)	Net buys (5)	Net sells (6)
Past return	-8.4501*** (-62.51)	7.9972*** (58.74)	-4.5860*** (-63.34)	3.9176*** (57.95)	-4.5070*** (-64.92)	3.9062*** (60.71)
Past turnover	-3.6916*** (-12.11)	-7.2845*** (-19.02)	-2.1964*** (-4.48)	1.5703*** (2.66)	3.8177*** (5.60)	6.5108*** (8.47)
Past volatility	46.8435*** (34.32)	-51.2077*** (-35.83)	35.4212*** (16.72)	-96.9425*** (-40.37)	15.8986*** (5.65)	-130.2595*** (-42.02)
Stock-month FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	11,256,384	11,999,824	11,256,394	11,999,835	11,254,789	11,997,894
adj. <i>R</i> ²	0.17	0.12	0.17	0.12	0.17	0.12

Panel (b): Dependent variable: Lambda

	Lagged 1-day		Lagged 5-days		Lagged 10-days	
	Net buys (1)	Net sells (2)	Net buys (3)	Net sells (4)	Net buys (5)	Net sells (6)
Past return	-0.1789*** (-27.30)	0.0315*** (5.31)	-0.1658*** (-45.13)	0.0468*** (15.15)	-0.1563*** (-50.97)	0.0517*** (19.04)
Past turnover	-0.6391*** (-33.05)	-0.7546*** (-36.16)	-0.6653*** (-20.69)	-0.9743*** (-27.08)	-0.2762*** (-6.49)	-0.7182*** (-15.29)
Past volatility	2.0497*** (26.13)	1.7077*** (23.44)	1.6103*** (12.42)	1.2327*** (9.48)	-0.0791 (-0.45)	-1.0507*** (-6.10)
Stock-month FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	11,643,212	12,481,578	11,639,822	12,476,825	11,632,858	12,466,883
adj. <i>R</i> ²	0.35	0.30	0.35	0.30	0.35	0.30

Panel (c): Dependent variable: Quote-based price impact

	Lagged 1-day		Lagged 5-days		Lagged 10-days	
	Net buys (1)	Net sells (2)	Net buys (3)	Net sells (4)	Net buys (5)	Net sells (6)
Past return	-0.2353*** (-23.32)	-0.0318*** (-3.33)	-0.2723*** (-49.77)	0.0778*** (15.28)	-0.2752*** (-59.69)	0.1085*** (23.93)
Past turnover	-1.3414*** (-36.70)	-1.2069*** (-32.36)	-1.9172*** (-32.59)	-1.9636*** (-32.63)	-1.7660*** (-22.29)	-1.9834*** (-24.08)
Past volatility	4.8702*** (37.80)	5.0263*** (39.82)	3.1059*** (14.31)	3.5527*** (16.95)	-0.5489* (-1.80)	-1.1362*** (-3.81)
Stock-month FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	11,632,436	12,446,048	11,629,070	12,441,313	11,622,111	12,431,403
adj. <i>R</i> ²	0.36	0.30	0.36	0.30	0.36	0.30

Panel (d): Ln(Amihud)

	Lagged 1-day		Lagged 5-days		Lagged 10-days	
	Net buys (1)	Net sells (2)	Net buys (3)	Net sells (4)	Net buys (5)	Net sells (6)
Past return	-2.2732*** (-86.84)	1.8995*** (64.44)	-1.2229*** (-94.90)	0.5658*** (45.06)	-1.0076*** (-96.74)	0.3665*** (34.90)
Past turnover	-7.3087*** (-72.03)	-10.0111*** (-86.49)	-6.4596*** (-46.66)	-8.4929*** (-51.85)	-3.6193*** (-20.10)	-5.2103*** (-24.02)
Past volatility	1.4582*** (6.93)	-4.4702*** (-18.28)	-9.8627*** (-27.07)	-17.7004*** (-41.81)	-21.1014*** (-39.52)	-31.8585*** (-50.58)
Stock-month FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	11,678,977	12,524,311	11,675,574	12,519,533	11,668,565	12,509,524
adj. <i>R</i> ²	0.69	0.63	0.70	0.63	0.70	0.63

5.4 Table 4: information environment

Read:

- Panel A shows that the skewness-return relation weakens after earnings announcements: the interaction coefficient is +0.5476.

- For price impact costs, the earnings interaction is +1.0181 on net-buy days and -0.9680 on net-sell days, muting the baseline asymmetry.
- Panel B shows similar attenuation for analyst coverage.
- Panel C shows similar attenuation for larger market capitalization.
- Panel D shows the opposite for complicated firms: firms with more business segments relative to size have stronger skewness and price-impact effects.

Economic message. UWIP is lower when public information and investor attention are better, and higher when firm information is complex. This supports the interpretation that the patterns come from uncertainty about what is already priced.

Table 4
Effect of the information environment.
Panel (a): Interaction with EA

	Return skewness (1)	Price impact costs		Lambda		Quote-based price impact		Ln(Amihud)	
		Net buys (2)	Net sells (3)	Net buys (4)	Net sells (5)	Net buys (6)	Net sells (7)	Net buys (8)	Net sells (9)
Past return	-2.0291*** (-66.88)	-2.9678*** (-55.24)	2.8325*** (52.72)	-0.0980*** (-40.09)	0.0254*** (11.00)	-0.2138*** (-50.73)	0.0734*** (16.81)	-0.8276*** (-85.04)	0.2658*** (26.70)
EA	-0.0296*** (-11.07)	-0.0331*** (-7.49)	0.0320*** (5.68)	-0.0019*** (-6.44)	-0.0012*** (-3.44)	-0.0005 (-0.85)	0.0006 (0.87)	-0.0226*** (-20.59)	-0.0177*** (-13.23)
Past return*EA	0.5476*** (21.54)	1.0181*** (20.81)	-0.9680*** (-17.23)	0.0317*** (10.29)	-0.0109*** (-3.39)	0.0474*** (8.14)	-0.0292*** (-4.63)	0.2302*** (22.29)	-0.1322*** (-11.99)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	19,386,514	9,497,462	9,655,532	9,754,434	9,956,693	9,749,027	9,935,083	9,781,451	9,987,895
adj. R ²	0.06	0.13	0.06	0.33	0.28	0.34	0.28	0.69	0.64

Panel (b): Interaction with AnalCov

	Return skewness (1)	Price impact costs		Lambda		Quote-based price impact		Ln(Amihud)	
		Net buys (2)	Net sells (3)	Net buys (4)	Net sells (5)	Net buys (6)	Net sells (7)	Net buys (8)	Net sells (9)
Past return	-3.1621*** (-74.15)	-8.9682*** (-73.67)	6.3611*** (58.33)	-0.3306*** (-54.21)	0.0802*** (14.83)	-0.4888*** (-55.46)	0.1579*** (19.20)	-1.7865*** (-115.26)	0.5756*** (33.47)
Past return*AnalCov	0.6929*** (41.69)	3.3782*** (59.87)	-2.1464*** (-40.82)	0.1334*** (47.12)	-0.0251*** (-9.23)	0.1633*** (40.26)	-0.0435*** (-10.80)	0.5969*** (85.59)	-0.1845*** (-24.09)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	23,900,236	11,254,789	11,997,894	11,632,858	12,466,883	11,622,111	12,431,403	11,668,565	12,509,524
adj. R ²	0.07	0.18	0.12	0.35	0.30	0.36	0.30	0.70	0.63

Panel (c): Interaction with Ln(mcap)

	Return skewness (1)	Price impact costs		Lambda		Quote-based price impact		Ln(Amihud)	
		Net buys (2)	Net sells (3)	Net buys (4)	Net sells (5)	Net buys (6)	Net sells (7)	Net buys (8)	Net sells (9)
Past return	-6.9808*** (-52.69)	-29.9299*** (-63.05)	19.9523*** (44.41)	-1.1031*** (-47.85)	0.2691*** (11.75)	-1.3889*** (-43.28)	0.4360*** (13.21)	-5.0139*** (-93.04)	1.8203*** (28.90)
Past return*Ln(mcap)	0.3926*** (40.05)	2.1085*** (57.31)	-1.3747*** (-39.21)	0.0785*** (44.97)	-0.0186*** (-10.44)	0.0923*** (38.27)	-0.0280*** (-10.98)	0.3321*** (81.83)	-0.1243*** (-26.05)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	23,814,464	11,235,102	11,975,676	11,589,146	12,411,244	11,578,495	12,376,175	11,623,981	12,452,720
adj. R ²	0.07	0.18	0.12	0.35	0.30	0.36	0.30	0.70	0.63

Panel (d): Interaction with #Segm^{orth}

	Return skewness (1)	Price impact costs		Lambda		Quote-based price impact		Ln(Amihud)	
		Net buys (2)	Net sells (3)	Net buys (4)	Net sells (5)	Net buys (6)	Net sells (7)	Net buys (8)	Net sells (9)
Past return	-2.2308*** (-71.15)	-4.3843*** (-60.29)	3.8170*** (56.82)	-0.1544*** (-46.90)	0.0520*** (17.97)	-0.2723*** (-55.85)	0.1078*** (22.62)	-0.9683*** (-88.87)	0.3550*** (32.42)
Past return*#Segm ^{orth}	-0.0781*** (-2.28)	-1.3792*** (-11.67)	1.0867*** (11.15)	-0.0527*** (-9.10)	0.0273*** (5.32)	-0.0611*** (-7.06)	0.0230*** (2.90)	-0.1378*** (-7.92)	0.1386*** (8.94)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	19,570,638	9,397,769	9,799,682	9,669,922	10,131,991	9,664,368	10,109,367	9,697,134	10,164,345
adj. R ²	0.06	0.17	0.12	0.37	0.32	0.37	0.31	0.70	0.64

This table reports results of cross-sectional tests for our return skewness and price impact regressions (see Section 3.2). For brevity, we only show results using lagged 10-day returns as the key independent variable. Panel (a) shows results for interacting the past return with an EA dummy, which flags whether or not there was a

5.5 Table 5 and Table 6: order-flow source and alternative explanations

Read: Table 5:

- Panel A splits by institutional order imbalance. Price impact costs have a coefficient of -3.3249 for institutional net buys and $+3.1638$ for institutional net sells.
- Lambda, quote-based price impact, and $\ln(\text{Amihud})$ show the same institutional buy-sell pattern.
- Panel B splits by retail order imbalance. The asymmetry is much weaker and often does not have the predicted sell-side sign.
- This supports the view that the mechanism is linked to informed or potentially informed trading.

Read: Table 6:

- Panel A shows that after S&P 500 additions, the UWIP footprint weakens.
- Panel B shows that after S&P 500 deletions, the footprint tends to strengthen.
- Panel C shows that short-selling fees do not strengthen the skewness-return relation or the price-impact asymmetry in the way short-sale-constraint theories predict.

Economic message. The retail and robustness tests make the UWIP interpretation more credible: the effect is strongest where adverse-selection concerns are relevant and does not line up with short-sale-constraint explanations.

Table 5
Institutional vs. retail trade imbalance.
Panel (a): Split based on institutional order imbalance

	Price impact costs		Lambda		Quote-based price impact		Ln(Amihud)	
	Net buys	Net sells	Net buys	Net sells	Net buys	Net sells	Net buys	Net sells
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Past return	-3.3249*** (-31.56)	3.1638*** (29.97)	-0.0732*** (-16.58)	0.0338*** (8.08)	-0.1369*** (-19.96)	0.0350*** (5.14)	-0.6850*** (-48.50)	0.0900*** (6.56)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	3,752,931	4,010,100	3,861,532	4,127,163	3,859,456	4,123,076	3,858,335	4,123,826
adj. R ²	0.10	0.07	0.17	0.16	0.31	0.28	0.78	0.77

	Price impact costs		Lambda		Quote-based price impact		Ln(Amihud)	
	Net buys	Net sells	Net buys	Net sells	Net buys	Net sells	Net buys	Net sells
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Past return	-0.4240*** (-7.81)	0.3101*** (5.85)	-0.0189*** (-6.27)	-0.0116*** (-3.76)	-0.0546*** (-9.52)	-0.0244*** (-4.24)	-0.3366*** (-27.07)	-0.2292*** (-18.85)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	3,243,393	3,772,296	3,340,080	3,872,364	3,339,477	3,871,691	3,339,856	3,872,166
adj. R ²	0.08	0.09	0.24	0.24	0.34	0.33	0.80	0.79

This table reports results of our price impact regressions, except that we now split the sample by institutional or retail trade imbalance instead of by (total) trade imbalance. See Section 3.3. For brevity, we only show results for the key independent variable. In Panel (a), we focus on stocks that are added to the S&P 500 index during our sample period. For each a we include 5 years before and after the addition event (while excluding the month of the addition event). We show results for interacting the past return with SP500 that takes the value one after the stock is added to the S&P 500 and zero before. SP500 is not added separately to the regression as it is subsumed by the stock-month FE. In Panel (b), we focus on stocks that are dropped from the S&P 500 index during our sample period (including 5 years before and after the deletion event). We show interacting the past return with *exSP500*, a dummy that takes the value one after the stock is dropped from the S&P 500 and zero before. *exSP500* is not added separately to the regression as it is subsumed by the stock-month FE. At the bottom of Panels (a) and (b), we show *t*-statistics for whether the coefficients of the interaction term with SP500 or *exSP500* are statistically significantly different between the subsamples of days with net buys and net sells. In Panel (c), we show results for the past return with *ShortFee*, defined as the average short selling fee (i.e., equity lending fee) for the stock over the prior month (data comes from Markit) and is one

Table 5: institutional vs. retail trade imbalance

Table 6
Alternative explanations.
Panel (a): S&P 500 additions (+/- 5 years)

	Price impact costs		Lambda		Quote-based price impact		Ln(Amihud)	
	Net buys	Net sells	Net buys	Net sells	Net buys	Net sells	Net buys	Net sells
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Past return	-1.2654*** (-21.05)	-0.1040*** (-6.10)	0.1952*** (3.77)	-0.0123*** (-5.53)	-0.0026 (-0.91)	-0.0991*** (-13.34)	0.0202*** (2.32)	-0.1599*** (-14.65)
Past return*SP500	0.4777*** (7.85)	0.0212 (0.79)	-0.0252 (-0.35)	0.0064*** (2.77)	0.0021 (0.68)	-0.0232*** (-2.35)	0.0068*** (6.26)	-0.0077 (-0.54)
t-statistic of diff. of interaction coef.		(0.58)		(1.21)		(3.25)***		(5.20)***
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	986,131	579,780	392,873	586,564	398,214	586,183	397,895	586,796
adj. R ²	0.05	0.08	0.02	0.29	0.22	0.46	0.41	0.77

	Price impact costs		Lambda		Quote-based price impact		Ln(Amihud)	
	Net buys	Net sells	Net buys	Net sells	Net buys	Net sells	Net buys	Net sells
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Past return	-1.0299*** (-14.71)	-0.1981*** (-4.86)	0.3071*** (3.39)	-0.0111*** (-4.00)	0.0003 (0.15)	-0.1111*** (-9.16)	-0.0074 (-0.50)	-0.1511*** (-8.32)
Past return*exSP500	-0.0018 (-0.02)	-0.3051*** (-4.31)	0.4857*** (2.90)	-0.0332*** (-3.42)	-0.0042 (-0.67)	-0.0409 (-1.48)	0.0273 (0.96)	-0.3063*** (-7.10)
t-statistic of diff. of interaction coef.		(-4.20)***		(-2.80)***		(-2.01)**		(-5.10)***
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	445,566	247,540	199,932	248,138	206,519	248,138	200,224	248,272
adj. R ²	0.04	0.08	0.00	0.43	0.36	0.45	0.38	0.59

	Price impact costs		Lambda		Quote-based price impact		Ln(Amihud)	
	Net buys	Net sells	Net buys	Net sells	Net buys	Net sells	Net buys	Net sells
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Past return	-0.9709*** (-37.22)	-1.6755*** (-33.83)	1.6487*** (32.16)	-0.0277*** (-0.94)	0.0274*** (6.89)	-0.1210*** (-17.84)	0.0267*** (4.65)	-0.6231*** (-14.21)
Past return*ShortFee	-0.0050 (-1.54)	-0.0020*** (-4.33)	-0.0343*** (-2.55)	-0.0026*** (-3.56)	-0.0011* (-1.79)	-0.0027*** (-2.11)	-0.0008 (-0.73)	-0.0227*** (-9.91)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	7,833,642	3,653,547	2,982,991	3,717,747	4,007,682	3,703,114	4,006,141	3,709,722
adj. R ²	0.08	0.10	0.06	0.16	0.15	0.31	0.28	0.78

This table reports results of additional cross-sectional tests for our return skewness and price impact regressions (see Section 3.3). For brevity, we only show results for the key independent variable. In Panel (a), we focus on stocks that are added to the S&P 500 index during our sample period. For each a we include 5 years before and after the addition event (while excluding the month of the addition event). We show results for interacting the past return with SP500 that takes the value one after the stock is added to the S&P 500 and zero before. SP500 is not added separately to the regression as it is subsumed by the stock-month FE. In Panel (b), we focus on stocks that are dropped from the S&P 500 index during our sample period (including 5 years before and after the deletion event). We show interacting the past return with *exSP500*, a dummy that takes the value one after the stock is dropped from the S&P 500 and zero before. *exSP500* is not added separately to the regression as it is subsumed by the stock-month FE. At the bottom of Panels (a) and (b), we show *t*-statistics for whether the coefficients of the interaction term with SP500 or *exSP500* are statistically significantly different between the subsamples of days with net buys and net sells. In Panel (c), we show results for the past return with *ShortFee*, defined as the average short selling fee (i.e., equity lending fee) for the stock over the prior month (data comes from Markit) and is one

Table 6: alternative explanations

6 Short summary

- The paper introduces uncertainty about what is in the price, or UWIP.
- UWIP means informed investors do not know whether their private signal is novel or already priced.
- Speculators use recent price movements to judge signal novelty.
- A positive signal after a price rise may be stale; a positive signal after a price decline is more likely novel.
- Market makers anticipate this behavior and make price impact asymmetric.
- After price increases, buy volume has lower price impact than sell volume.
- After price decreases, sell volume has lower price impact than buy volume.
- The model predicts that return skewness is negatively related to past returns.
- It also predicts that price impact falls with past returns for buys and rises with past returns for sells.
- The empirical analysis uses NYSE stocks from 1993 to 2014 and intraday TAQ data.
- Realized daily skewness is negatively related to lagged returns.
- Price-impact regressions show opposite signs for net-buy and net-sell days.
- These effects are weaker after earnings announcements, for firms with more analysts, and for large firms.
- The effects are stronger for complicated firms with many business segments relative to size.
- The asymmetry is much stronger for institutional order imbalance than retail order imbalance.
- The results are not well explained by inventory risk, S&P 500 inclusion effects, or short-sale constraints.
- UWIP lowers price informativeness because speculators trade less aggressively when they fear their information is stale.
- The broad takeaway is that the content of prices is itself uncertain, and that uncertainty shapes trading and price dynamics.

7 Conclusion

7.1 Core conclusion

Peress and Schmidt show that investors face uncertainty not only about fundamentals but also about what information current prices already contain. This uncertainty changes how speculators trade and how market makers price order flow. The main footprint is an asymmetric response of price impact to past returns: after price increases, buys are less informative than sells; after price decreases, sells are less informative than buys.

7.2 Economic intuition

The intuition is simple. A speculator with positive information wants to know whether the market already learned the same information. If the price recently increased, the answer may be yes, so she trades less aggressively. If the price recently decreased, her positive information is more likely to be new, so she trades more aggressively. Market makers understand this inference and adjust price impact accordingly. This produces negative dependence of skewness on past returns and asymmetric price impact across buy and sell order flow.

7.3 Takeaway

The paper adds a new informational friction to market microstructure: uncertainty about the content of prices. It explains why past price movements affect the informativeness of current order flow and why return distributions become skewed in a direction tied to prior returns. The evidence suggests that this uncertainty is a real concern for stock market participants, especially in more opaque and complex information environments.